Prob 1

a) $f(t) = 1 + 2t - 4t^4 \Rightarrow \underline{F(s) = \frac{1}{s} + \frac{2}{s^2} - \frac{96}{s^5}}$

b) $f(t) = 2t^2 e^{3t} \Rightarrow F(s) = \frac{4}{(s-3)^3}$

c) $f(t) = e^{-2t}\sin(5t) \Rightarrow F(s) = \frac{5}{(s+2)^2 + 25}$

Prob 2

a) $F(s) = \frac{3}{s} + \frac{1}{s^8} \Rightarrow f(t) = 3 + \frac{t^7}{7!}$

b) $F(s) = \frac{2s+1}{s^2 - s - 2} = \frac{2s+1}{(s-2)(s+1)} = \frac{5}{3(s-2)} + \frac{1}{3(s+1)} \Rightarrow f(t) = \frac{1}{3}(5e^{2t} + e^{-t})$

$2s + 1 = As + A + Bs - 2B \quad A = 5/3,\ B = 1/3$

c) $F(s) = \frac{1}{(s+1)(s^2+1)} = \frac{1}{2(s+1)} - \frac{s}{2(s^2+1)} + \frac{1}{2(s^2+1)} \Rightarrow f(t) = \frac{1}{2}(e^{-t} - \cos(t) + \sin(t))$

$\frac{1}{(s+1)(s^2+1)} = \frac{A}{(s+1)} + \frac{Bs+C}{(s^2+1)} \Rightarrow 1 = As^2 + A + Bs^2 + Cs + Bs + C \Rightarrow A = C = 1/2,\ B = -1/2$

Prob 4 - IVP

a) $2y'' + y' - y = 0,\ y(0) = 0,\ y(0) = 1$

$2(s^2F(s) + s\overset{0}{f(0)} + f'(0)) + (sF(s) + \overset{0}{f(0)}) - F(s) = 0$

$2s^2F(s) + 2 + sF(s) - F(s) = 0$

$F(s)(2s^2 + s - 1) = -2$

$F(s) = \frac{-2}{(2s+1)(s-1)} = \frac{4}{6(s+1/2)} - \frac{2}{3(s-1)} \Rightarrow f(t) = \frac{2}{3}(e^{\frac{-t}{2}} - e^{t})$

$-2 = A(s-1) + B(2s+1) \Rightarrow -2 = As - A + 2Bs + B,\ A = 4/3,\ B = -2/3$